

NATURE DEPRESSES THE CUBIC

Cardano, Weyl curvature, and an equilateral spectral locus in Petrov type I

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Abstract

Depressing a cubic is the translation that carries the centroid of its roots to the origin, and the Weyl tensor arrives with that translation already applied by the trace-removal separating Weyl curvature from the Ricci-determined part of Riemann. Because the self-dual Weyl operator Q is a symmetric tracefree complex 3×3 matrix, its characteristic polynomial is already in Cardano's depressed form $\lambda^3 + p\lambda + q$, and its three eigenvalues therefore form a triangle in $\mathbb{C}$ whose centroid is pinned at the origin. This note identifies the locus on which that triangle is equilateral. Assuming $J \neq 0$, so that the spectrum does not collapse,

$$\text{equilateral spectrum} \iff p = 0 \iff I = 0 \iff \text{tr}(E^2) = \text{tr}(H^2) \text{ and } \text{tr}(EH) = 0,$$

so that the vanishing of the quadratic Weyl invariant corresponds to equal electric and magnetic contributions to the Bel–Robinson super-energy density together with the trace-orthogonality of the two parts. The curvature nevertheless remains algebraically general, so this is a real codimension-two locus inside Petrov type I. On it $\text{spec } Q$ is a scaled copy of $\mu_3 = \{1, \omega, \omega^2\}$, so the order-three symmetry implicit in Cardano's radical parametrisation becomes an actual symmetry of the curvature spectrum. The Petrov quartic, this characteristic cubic and its Marden derivative moreover form a $4 \rightarrow 3 \rightarrow 2$ algebraic cascade, the cubic being the classical resolvent of the Weyl quartic. The resulting Eisenstein recognition is therefore structural and cyclotomic, implying no integral lattice structure on spacetime.

What have cubics ever done for us?

School mathematics spends years on quadratics and remarkably little time on what comes next. Cubics tend to appear, if at all, as equations whose exact solution is too cumbersome to be useful, while quartics barely appear at all. Yet historically the degrees are entangled. Quartics were solved by reducing their difficulty to a resolvent cubic; cubics brought with them complex numbers and roots of unity even when the problem began over the reals.

In general relativity the same hierarchy appears unexpectedly in the algebra of curvature. The Petrov description of the Weyl tensor gives a quartic in the principal null directions; its classical resolvent is the characteristic cubic of the self-dual Weyl operator; and the derivative of that cubic is the quadratic whose roots, by Marden's theorem, locate the foci of the Steiner inellipse of its spectrum. Thus

$$4 \longrightarrow 3 \longrightarrow 2$$

is not a march towards greater complication, but a descent through different descriptions of the same structure.

1 THE DEPRESSION THAT WAS ALREADY DONE

Cardano's method for the cubic begins with an act of tidying. Given $x^3 + ax^2 + bx + c$ one substitutes $x \mapsto x - a/3$, which translates the root configuration in $\mathbb{C}$ until the roots sum to zero and the quadratic term disappears; the resulting *depressed* cubic $\lambda^3 + p\lambda + q$ is then attacked by $\lambda = u + v$ with $3uv = -p$. Students meet the depression as a manoeuvre and quite reasonably experience it as arbitrary, a change of variable performed because it happens to work. Stated geometrically it is less arbitrary and more memorable, since the substitution is exactly the translation carrying the centroid of the roots to the origin; it is the degree-one case of a Tschirnhaus transformation, and we shall refer to it informally as *centring the spectrum*, meaning nothing by that beyond the requirement that the roots sum to zero.

There is at least one place in mathematical physics where that centring has already been done by the geometry. The Weyl tensor is by construction the tracefree part of the Riemann tensor, its traces having been removed into the Ricci tensor; read as a linear operator on the three-dimensional space of self-dual bivectors, that tracefreeness descends to the operator, and the characteristic polynomial therefore has no quadratic term. The sum of the three curvature eigenvalues vanishes because the Ricci-determined part of Riemann has been separated off. Weyl tracefreeness is Cardano depression already performed, and it is performed for reasons that have nothing to do with the convenience of solving anything.

Once the depressed form is taken seriously as a geometric fact, the classical apparatus attached to it becomes available for reading physically, and this note follows one thread of that to its end. The thread is the locus on which the root triangle is equilateral, which turns out to have a clean invariant characterisation in terms of the electric and magnetic parts of the Weyl tensor. The ingredients are old and the bibliography records where each belongs; what is offered is the arrangement, together with the remark that it lets Cardano's depression be motivated by pointing at the sky.

2 THE SELF-DUAL WEYL OPERATOR

On a four-dimensional Lorentzian manifold the bundle Λ^2 has rank six and the Hodge star acting on it satisfies $\star^2 = -1$, so its eigenvalues are $\pm i$ and the projectors $P^\pm = \frac{1}{2}(1 \mp i\star)$ split the complexified bundle as $\Lambda^2 \otimes \mathbb{C} = \Lambda^+ \oplus \Lambda^-$, each summand of complex rank three. The Weyl tensor, viewed as a symmetric operator on Λ^2 , commutes with $\star$ and therefore preserves the splitting, acting on Λ^+ as a complex-linear operator Q ; for a real Lorentzian Weyl tensor the anti-self-dual sector is the complex conjugate of the self-dual one, so Q contains the independent information. In an orthonormal basis of Λ^+ this Q is a symmetric 3×3 complex matrix with

$$\text{tr } Q = 0. \quad (1)$$

This is Petrov's own presentation of the classification, in which the algebraic types are read from the eigenvalue and canonical (Jordan) structure of precisely this operator [1, 2]. In the chiral first-order formulations of general relativity, most transparently Plebanski's, the same object is the field Ψ_{ij} surviving the simplicity constraint, its tracefree part being the self-dual Weyl operator and its trace carrying Λ [11, 12].

Relative to any unit timelike u^a the Weyl tensor decomposes into electric and magnetic parts E_{ab} and H_{ab} , each real, symmetric and tracefree on the orthogonal three-space, and

$$Q = E + iH \quad (2)$$

Convention	Definitions	Speciality condition
Spinorial (Penrose–Rindler)	$I = \frac{1}{2}\Psi_{ABCD}\Psi^{ABCD}$, $J = \frac{1}{6}\Psi_{AB}{}^{CD}\Psi_{CD}{}^{EF}\Psi_{EF}{}^{AB}$	$I^3 = 27J^2$
Matrix traces	$I_Q = \text{tr}(Q^2) = a$, $J_Q = \text{tr}(Q^3) = b$	$I_Q^3 = 6J_Q^2$

Table 1: Two normalisations of the Weyl invariants, and the corresponding form of the relation characterising algebraic speciality. The famous 27 belongs to the spinorial normalisation and the raw matrix traces give 6; the two rows are related by $a = 2I$ and $b = 6J$, derived in Proposition 8. Compare [5, 7, 4].

is that operator written in an adapted frame [7]. The eigenvalues of Q are observer-independent although E and H separately are not. Readers who would like the passage from the four-index tensor C_{abcd} to the 3×3 matrix Q set out rather than asserted, together with its relation to the more familiar spinorial route through principal null directions, will find it in Appendix A.

3 THE CHARACTERISTIC CUBIC

Because Q is tracefree,

$$\det(\lambda \mathbb{1} - Q) = \lambda^3 - \frac{1}{2} \text{tr}(Q^2) \lambda - \det Q = \lambda^3 + p\lambda + q, \quad (3)$$

with $p = -\frac{1}{2} \text{tr}(Q^2)$ and $q = -\det Q = -\frac{1}{3} \text{tr}(Q^3)$, the last equality being the Newton identity for a tracefree 3×3 matrix. Writing $a = \text{tr}(Q^2)$ and $b = \text{tr}(Q^3)$ the characteristic equation reads $\lambda^3 - \frac{1}{2}a\lambda - \frac{1}{3}b = 0$, which is the form in which Ferrando and Sáez state it [4].

The two coefficients are the two independent scalar invariants of the Weyl tensor at a point, usually written I and J . Their normalisation is not standardised, and a good deal of apparent disagreement in the literature is a matter of where the halves and sixths have been parked, so Table 1 records the two common choices. The relation displayed in its final column is a *syzygy* in the classical sense, a polynomial identity among invariants that holds on a subvariety rather than an equation to be solved: I and J are independent on a generic spacetime, and the algebraically special ones are exactly those where they become dependent. We use p and q throughout, and write $I = 0$ for the normalisation-independent statement $\text{tr}(Q^2) = 0$.

Proposition 1 (Discriminant). $\text{disc} = -4p^3 - 27q^2 = [(\lambda_1 - \lambda_2)(\lambda_1 - \lambda_3)(\lambda_2 - \lambda_3)]^2 = \frac{1}{2}(a^3 - 6b^2)$, and the Weyl tensor has a repeated eigenvalue if and only if $\text{disc} = 0$.

The first coarse division of Petrov type is encoded by the root multiplicities of this cubic, with the Jordan structure supplying the remaining refinement: vanishing discriminant separates type I from the rest, while type D is diagonalisable on a characteristic polynomial shared with type II, and III is separated from N by the degree of the minimal polynomial [4, 2]. Characteristic data and operator structure are genuinely different layers, and the cubic sees only the first.

4 THE ROOT TRIANGLE

Regard the three eigenvalues as points of $\mathbb{C}$. By (1) their sum vanishes, so the centroid of the triangle sits at the origin; this is the geometric content of the depression, and general relativity has applied it by the trace-removal that separates Weyl curvature from the Ricci-determined part of Riemann.

Petrov type	Root pattern	Root triangle	Invariants
I (general)	three distinct roots	nondegenerate	$\text{disc} \neq 0$
I, equilateral locus	$\mu, \omega\mu, \omega^2\mu$	equilateral	$p = 0, q \neq 0$
II, D	one double root	degenerate to a segment	$\text{disc} = 0, q \neq 0$
III, N	triple root 0, nilpotent	collapsed to a point	$p = q = 0, Q \neq 0$
O	$Q = 0$	no triangle	$Q = 0$

Table 2: Petrov types as root configurations. The minimal polynomial distinguishes II from D and III from N; the characteristic cubic alone does not.

Cardano's substitution $\lambda = u + v$ with $3uv = -p$ gives $u^3 + v^3 = -q$ and $u^3v^3 = -p^3/27$, whence

$$\lambda_1 = u + v, \quad \lambda_2 = \omega u + \omega^2 v, \quad \lambda_3 = \omega^2 u + \omega v, \quad \omega = e^{2\pi i/3}. \quad (4)$$

Whenever the radicand is nonzero the choice of cube root is a μ_3 -torsor, and the threefold ambiguity appears as the cyclic action of μ_3 on the roots; this is the cyclotomic content of the radical solution, and it lives in the parametrisation rather than in the spectrum. What varies from one cubic to the next is whether μ_3 ever becomes visible in the answer, and §5 identifies the condition under which it does.

Proposition 2 (Equilaterality). The triangle is equilateral if and only if $p = 0$, that is, if and only if $\text{tr}(Q^2) = 0$.

Proof. A triangle with centroid at the origin is equilateral exactly when its vertices form a μ_3 -orbit, so after relabelling $\lambda_2 = \omega\lambda_1$ and $\lambda_3 = \omega^2\lambda_1$; then $p = \sum_{i<j} \lambda_i\lambda_j = \lambda_1^2(\omega + \omega^2 + 1) = 0$. Conversely if $p = 0$ the cubic reads $\lambda^3 = -q$, whose roots are the three cube roots of $-q$. $\square$

Remark 3 (Marden's reading). Marden's theorem states that the foci of the Steiner inellipse of the triangle of roots of a cubic are the roots of its derivative [16, 17]. Here the derivative is $3\lambda^2 + p$, with roots $\pm\sqrt{-p/3}$, and these coincide precisely when $p = 0$. An ellipse with coincident foci is a circle, and a triangle whose Steiner inellipse is a circle is equilateral. The chain is worth displaying, since it gives the same locus a second and entirely geometric reading:

$$\begin{aligned} I = 0 &\iff p = 0 \iff \chi'(\lambda) = 3\lambda^2 \\ &\iff \text{the critical points coalesce at the origin} \\ &\iff \text{the Steiner inellipse is a circle.} \end{aligned}$$

This is why the condition deserves a picture.

5 THE EQUILATERAL LOCUS

We can now state the result the note exists for.

Theorem 4. Let $Q = E + iH$ be the self-dual Weyl operator at a point, and suppose $q \neq 0$, equivalently $J \neq 0$. Then the following are equivalent:

- (i) the spectrum of Q is an equilateral triangle in $\mathbb{C}$;
- (ii) $p = 0$, that is $I = 0$;
- (iii) $\text{tr}(E^2) = \text{tr}(H^2)$ and $\text{tr}(EH) = 0$.

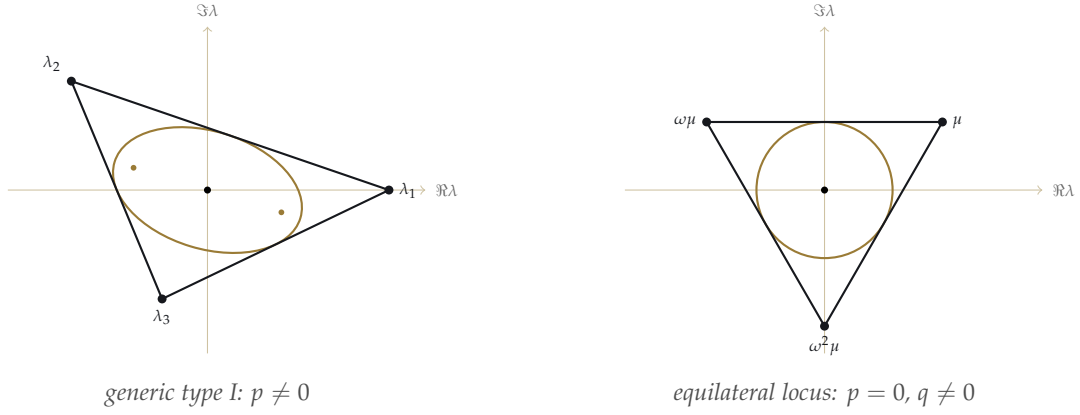


Figure 1: The Weyl spectrum as a triangle with centroid pinned at the origin by tracefreeness, shown with its Steiner inellipse. The foci of that inellipse are the roots of $3\lambda^2 + p$; they merge at the origin, and the ellipse becomes a circle, exactly when $p = 0$.

When they hold, $\text{spec } Q = \mu \cdot \mu_3$ for some $\mu \neq 0$ and the Weyl tensor is of Petrov type I. The hypothesis $q \neq 0$ is not cosmetic: $p = q = 0$ describes the nilpotent operators of types III and N, whose three eigenvalues coincide at the origin, and a triple point is not a triangle.

Proof. (i) $\Leftrightarrow$ (ii) is Proposition 2. For (ii) $\Leftrightarrow$ (iii), E and H are real symmetric, so $\text{tr}(Q^2) = \text{tr}(E^2) - \text{tr}(H^2) + 2i \text{tr}(EH)$, and a complex number vanishes exactly when its real and imaginary parts do. For the last clause, $p = 0$ reduces the cubic to $\lambda^3 = -q$, whose roots are $\mu, \omega\mu, \omega^2\mu$ with $\mu^3 = -q \neq 0$; and $\text{disc} = -27q^2 \neq 0$, so the three are distinct and the type is I. $\square$

The physical reading of (iii) is that the electric and magnetic parts contribute equally to the Bel–Robinson super-energy density, whose density is built from $\text{tr}(E^2) + \text{tr}(H^2)$, and that the two tensors are trace-orthogonal; this is what gives the vanishing of the quadratic invariant a meaning an observer could in principle measure.

Corollary 5. A spacetime on this locus admits no observer for whom the Weyl tensor is purely electric or purely magnetic.

Proof. If $H = 0$ then $\text{tr}(Q^2) = \text{tr}(E^2) = \sum_{ij} E_{ij}^2 \geq 0$ with equality only for $E = 0$, so $p = 0$ forces $Q = 0$; symmetrically for $E = 0$. This is consistent with the criterion of McIntosh, Arianrhod, Wade and Hoenselaers, in which the existence of such an observer requires I real and nonzero, the sign distinguishing the electric from the magnetic case [10]. $\square$

Remark 6 (Placement inside type I). Since $I = J = 0$ characterises types III, N and O [6], and here I vanishes while J does not, the locus lies inside the algebraically general type. In the refinement of type I by $M = I^3/J^2$ introduced by Arianrhod and McIntosh [8, 9, 10], the algebraically special locus sits at the value of M fixed by that syzygy and the present locus sits at $M = 0$. Condition (iii) is two real equations, so one should expect it on a codimension-two subset of a generic spacetime.

Remark 7 (Why the Maxwell analogue behaves oppositely). Condition (iii) has the shape of the null condition for a Maxwell field, $|E|^2 = |B|^2$ with $E \cdot B = 0$, one level up in tensor rank. The conclusions nevertheless differ, since a null Maxwell field is type N while Theorem 4 leaves us in type I, and the reason is a count of invariants. The Maxwell spinor φ_{AB} has the

single invariant $\varphi_{AB}\varphi^{AB}$, which is the discriminant of the binary quadratic $\varphi_{AB}\xi^A\xi^B$ whose roots are the two principal null directions, so its vanishing is degeneracy by definition. The Weyl spinor has two independent invariants, and the contrast is visible almost without prose:

	Maxwell	Weyl
spinor	φ_{AB}	Ψ_{ABCD}
principal polynomial	quadratic	quartic
independent basic invariants	1	2
null or special condition	$I = 0$	$I^3 - 27J^2 = 0$

For Maxwell $I = 0$ is the whole of the degeneracy condition; for Weyl it is one equation among the two whose combination controls degeneracy, and by itself it is transverse to it.

The Eisenstein recognition, and its limits

The three points of μ_3 form an equilateral triangle inside the triangular lattice $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$, whose full unit group $\mu_6 = \mathbb{Z}[\omega]^\times$ generates its sixfold rotational symmetry and whose Voronoi cell is a regular hexagon. That is a recognition of shape, and it is worth stating plainly what it is not. The scale μ is an arbitrary complex number, the eigenvalues are continuous functions of position with no reason to lie in a lattice, and nothing in Theorem 4 makes norms, ideals, prime splitting or cubic residues operative. The μ_3 occurring here is cyclotomic and Galois-theoretic in origin, arriving from the cube root in Cardano's formula; no integral Eisenstein structure has been placed on spacetime.

The appearance of μ_3 is nevertheless more than a pictorial coincidence, since it is the finite multiplicative symmetry forced by the cubic itself. Cardano introduces it through the ambiguity of extracting a cube root, where it acts on the parametrisation (u, v) of (4) as a redundancy of the radical parametrisation rather than as a multiplicative symmetry of the root set; on the locus $p = 0$ that same action descends to the spectrum, and

$$\omega \cdot \text{spec } Q = \text{spec } Q. \quad (5)$$

For a generic cubic μ_3 is a symmetry of Cardano's parametrisation; at $p = 0$ it becomes a symmetry of the spectrum. That is the payoff of the depression, and it is the reason the Eisenstein lattice is the right picture to have in mind even though none of its arithmetic is in play: the order-three rotation organising Cardano's radicals is precisely the rotation underlying the triangular lattice, and at $I = 0$ it is an honest symmetry of the curvature eigenvalues.

One structural point does deserve emphasis, because the natural first guess is to look for ω in the wrong place. Since $\star^2 = -1$, the minimal polynomial of the star operator is $t^2 + 1$ and its eigenvalues are $\pm i$, so the metric-induced Hodge structure supplies a two-sector chiral decomposition, and the μ_3 symmetry found above cannot arise as a third Hodge eigenspace. The order-three symmetry identified above accordingly belongs to the spectrum of the Weyl endomorphism and not to any grading of the bivector bundle, and keeping those two layers separate is what allows the recognition to be stated without overclaiming.

6 OUTLOOK

Three classical polynomials of descending degree have appeared, each attached to a different aspect of the same curvature: the binary quartic $\Psi_{ABCD}\xi^A\xi^B\xi^C\xi^D$, whose roots are the principal null directions; the characteristic cubic $\lambda^3 + p\lambda + q$, whose roots are the Weyl eigenvalues; and its derivative $3\lambda^2 + p$, whose roots are the foci of the Steiner inellipse of those eigenvalues.

The locus studied here is exactly where the last of these degenerates, $3\lambda^2 + p \rightarrow 3\lambda^2$, the two foci colliding at the origin; and by Proposition 8 the passage from the first to the second is the classical resolvent, so the cascade $4 \rightarrow 3 \rightarrow 2$ is a chain of algebraic relations rather than a sequence of analogies. Two questions remain open.

1. Which exact solutions, if any, satisfy $\text{tr}(E^2) = \text{tr}(H^2)$ and $\text{tr}(EH) = 0$ on an open set? In the space of Weyl tensors the condition is real codimension two, so under $x \mapsto Q(x)$ and generic transversality one expects its zero set in a four-dimensional spacetime to be a surface; a solution carrying it on an open set would therefore be badly non-transverse, and both the solution and its symmetry group would be worth knowing.
2. Does the Cardano pair (u, v) of (4) admit an invariant geometric interpretation for the Weyl tensor, or are u and v merely the coordinates in which the resolvent factorises?

Whether an integral or discrete geometric setting can promote this cyclotomic spectral configuration into genuine Eisenstein arithmetic is a separate question, and a separate paper.

A FROM THE WEYL TENSOR TO PETROV CUBIC: THREE VIEWS OF THE SAME CURVATURE

The main text moves briskly from C_{abcd} to a complex 3×3 matrix. For a relativist that step is routine; for the reader this note is otherwise aimed at, it is exactly the bridge that needs building, and the purpose of this appendix is to build it and then to say precisely how the cubic studied above is related to the quartic through which Petrov classification is usually first encountered.

A.1 The Weyl tensor as an operator on bivectors

In its usual tensorial presentation the Weyl curvature at an event is the real fourth-rank tensor C_{abcd} . Its antisymmetry in each index pair allows it equally to be regarded as an endomorphism of the six-dimensional real space of bivectors,

$$C : \Lambda^2 \longrightarrow \Lambda^2, \quad X_{ab} \longmapsto \frac{1}{2} C_{ab}{}^{cd} X_{cd}.$$

This change of viewpoint is the first step towards the Petrov classification. Rather than asking for the individual components of C_{abcd} , we ask how the curvature operator acts on the oriented two-planes through the event.

In Lorentzian four-space the Hodge operator satisfies $\star^2 = -1$ on Λ^2 , so after complexification its eigenvalues are $\pm i$ and

$$\Lambda^2 \otimes \mathbb{C} = \Lambda^+ \oplus \Lambda^-, \quad P^\pm = \frac{1}{2}(1 \mp i\star).$$

The Weyl operator commutes with $\star$, so each chiral subspace is preserved, and since Λ^+ has complex dimension three the self-dual Weyl curvature becomes

$$Q = C^+|_{\Lambda^+} : \Lambda^+ \longrightarrow \Lambda^+,$$

represented in a suitable basis by a complex symmetric tracefree 3×3 matrix. That the self-dual two-forms deserve to be treated as primary objects rather than as a convenient rewriting is the content of the chiral first-order formulations, in which Σ^{AB} and the self-dual curvature are the fundamental variables [11, 12, 15].

A.2 The spinor picture, and why Petrov is usually a quartic

Under the spinor correspondence the Weyl tensor becomes

$$C_{abcd} \longleftrightarrow \Psi_{ABCD} \epsilon_{A'B'} \epsilon_{C'D'} + \bar{\Psi}_{A'B'C'D'} \epsilon_{AB} \epsilon_{CD},$$

with all the self-dual information carried by the totally symmetric Weyl spinor $\Psi_{ABCD} = \Psi_{(ABCD)}$ [3]. Contracting with a projective spinor gives

$$\Psi_{ABCD} \xi^A \xi^B \xi^C \xi^D = 0,$$

a homogeneous binary quartic. Choosing a spin frame and writing the projective spinor as $\xi^A \sim (1, z)$ turns it into an ordinary quartic polynomial in z , whose four roots determine the principal spinors and hence the principal null directions; the Petrov types are their coincidence patterns:

$$\text{I} : 1+1+1+1, \quad \text{II} : 2+1+1, \quad \text{D} : 2+2, \quad \text{III} : 3+1, \quad \text{N} : 4, \quad \text{O} : \Psi_{ABCD} = 0.$$

It is worth saying plainly that *Petrov's quartic of principal null directions is not Cardano's cubic of Weyl eigenvalues*. They are different algebraic windows onto the same curvature, of different degrees, and confusing them is the natural trap this appendix exists to close:

C_{abcd}	$\longleftrightarrow$	Ψ_{ABCD}
$\downarrow$		$\downarrow$
$Q : \Lambda^+ \rightarrow \Lambda^+$		$\Psi_{ABCD} \xi^A \xi^B \xi^C \xi^D$
cubic eigenvalue equation		quartic PND equation

The relation between the two sides is closer than mere coexistence, and it can be made exact rather than argued by analogy. A binary quartic $f(\xi) = \sum_k \binom{4}{k} a_k \xi_0^{4-k} \xi_1^k$ carries two classical invariants,

$$I = a_0 a_4 - 4a_1 a_3 + 3a_2^2, \quad J = \det \begin{pmatrix} a_0 & a_1 & a_2 \\ a_1 & a_2 & a_3 \\ a_2 & a_3 & a_4 \end{pmatrix}, \quad (6)$$

of degrees two and three, its discriminant is proportional to $I^3 - 27J^2$, and its resolvent cubic is $4z^3 - Iz - J = 0$, the Weierstrass form. Setting $a_k = \Psi_k$ makes these the invariants of the Weyl quartic, and they are exactly the invariants of the spinorial row of Table 1; the 27 in the speciality syzygy is therefore the 27 of the Weierstrass discriminant, and the 6 in the matrix-trace row follows from Proposition 8 below rather than being an independent convention.

Proposition 8. Let Q act on symmetric spinors by $(Q\varphi)_{AB} = \Psi_{AB}{}^{CD} \varphi_{CD}$, and let I and J be as in (6) for the Weyl quartic. Then

$$\text{tr } Q = 0, \quad \text{tr}(Q^2) = 2I, \quad \text{tr}(Q^3) = 6J,$$

so that $\chi_Q(\lambda) = \lambda^3 - I\lambda - 2J$; and the substitution $\lambda = 2z$ carries χ_Q into the classical resolvent cubic $4z^3 - Iz - J$. The eigenvalues of the self-dual Weyl operator are therefore exactly twice the Weierstrass roots of the Weyl quartic.

Proof. The three trace identities are polynomial identities in $\Psi_0, \dots, \Psi_4$ of degrees one, two and three, verified by direct contraction in a spin dyad. Given them, (3) reads $\lambda^3 - \frac{1}{2}(2I)\lambda - \frac{1}{3}(6J)$, and substituting $\lambda = 2z$ gives $8z^3 - 2Iz - 2J = 2(4z^3 - Iz - J)$. $\square$

The scale factor is thus 2 and not a matter of convention, and the type D case confirms it exactly rather than up to an unexplained half. With only Ψ_2 nonvanishing (6) gives $I = 3\Psi_2^2$ and $J = -\Psi_2^3$, so the resolvent $4z^3 - 3\Psi_2^2 z + \Psi_2^3 = 0$ has roots $\{\Psi_2/2, \Psi_2/2, -\Psi_2\}$, and doubling returns the familiar Weyl eigenvalue triple $\{\Psi_2, \Psi_2, -2\Psi_2\}$ of Schwarzschild and Kerr. Consistency of the two rows of Table 1 also follows, since $a = \text{tr}(Q^2) = 2I$ and $b = \text{tr}(Q^3) = 6J$ turn $I^3 = 27J^2$ into $a^3 = 6b^2$.

This also explains, rather than merely warns of, the loss of information noted after Proposition 1. A resolvent cubic records how the four roots of the quartic pair up, which is why it distinguishes type I from the rest and why it cannot separate a diagonalisable double eigenvalue from a Jordan block; the eigendirections live in the quartic and in the operator, and not in the characteristic polynomial of either.

A.3 What an observer sees

Choosing a unit timelike observer u^a decomposes the Weyl tensor into its electric and magnetic parts,

$$E_{ab} = C_{acbd}u^c u^d, \quad H_{ab} = {}^*C_{acbd}u^c u^d,$$

both real, symmetric and tracefree on the orthogonal three-space, and in the self-dual representation they combine as $Q = E + iH$. The three descriptions used in this note,

$$C_{abcd}, \quad \Psi_{ABCD}, \quad Q = E + iH,$$

are therefore tensorial, spinorial and observer-adapted presentations of one curvature, and Theorem 4 is a statement about all three at once. The quadratic combinations $\text{tr}(E^2)$ and $\text{tr}(H^2)$ appearing in it are close relatives of the components of the Bel–Robinson tensor, whose super-energy density is essentially $\text{tr}(E^2) + \text{tr}(H^2)$, so that condition (iii) states that the two parts contribute equally to it; the formal parallel with the Maxwell stress-energy tensor is developed in [14, 13].

B A MINIMAL COMPUTATIONAL HANDLE

For readers who want to experiment with the locus rather than check the algebra, the following suffices to generate points on it and to confirm the discriminant identity of Proposition 1.

```

1 import numpy as np
2 w = np.exp(2j*np.pi/3)
3
4 def spectrum_data(Q):
5     # Q symmetric, traceless, complex 3
6     a, b = np.trace(Q@Q), np.trace(Q@Q@Q)
7     p, q = -a/2, -b/3
8     return p, q, -4*p**3 - 27*q**2, (a**3 - 6*b**2)/2 # last two agree
9
10 D = np.diag([mu := 0.83-1.4j, mu*w, mu*w**2]) # a point on the
11     locus
12 E, H = D.real, D.imag
13 print(np.trace(D@D), np.trace(E@E) - np.trace(H@H), np.trace(E@H)) # all ~
14     0

```

Any complex-orthogonal conjugate of D is another point on the locus with the same spectrum; D itself already realises it, so no conjugation is needed to exhibit an example.

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